

Project Report: Office Network Configuration

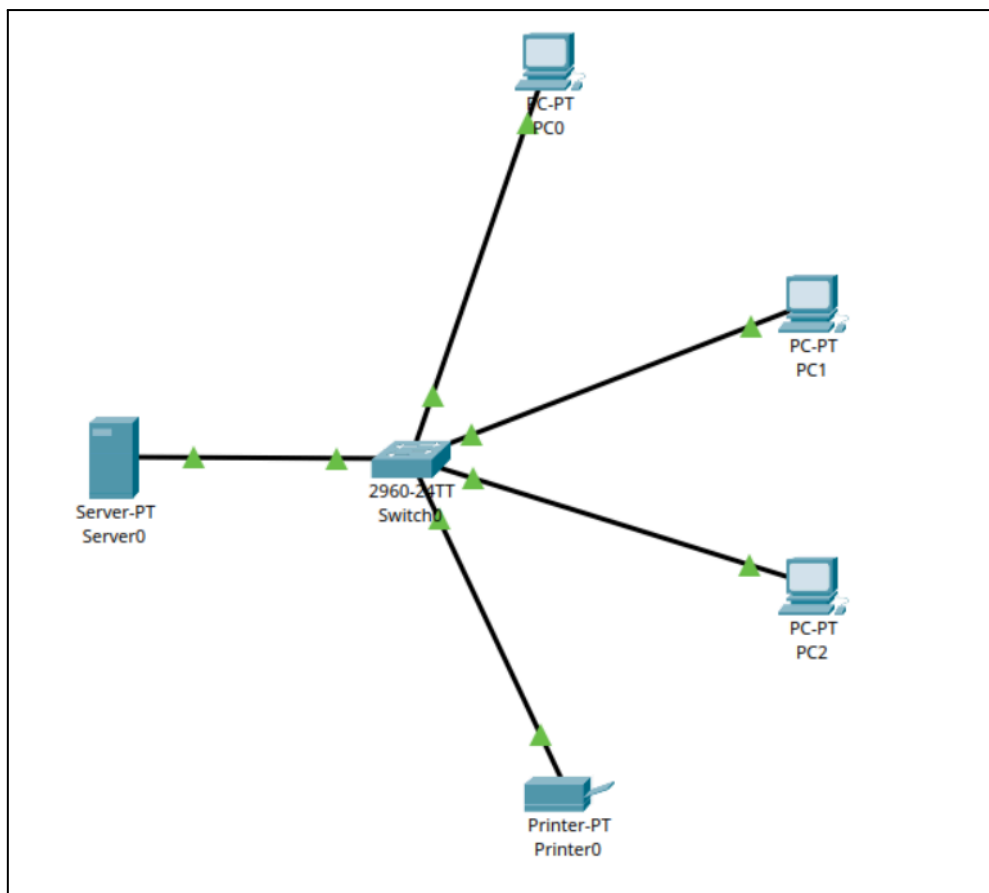
1. Project Overview & Repository

The objective of this project is to simulate and document a small office network infrastructure. This setup includes centralized file storage, network printing, local domain resolution, and a robust automated backup strategy to ensure data resilience.

- **GitHub Repository:** https://github.com/Leo-PHAV/EUTech_Leo-PHAV.git
- **Branch:** [task-office-network-Leo-PHAV](#)

2. Methodology & Technical Stack

- **Simulation Tool:** Cisco Packet Tracer used for network topology and service simulation.
- **Network Nodes:** 1 Server (Files/DNS), 3 PC Clients, and 1 Network Printer.
- **Addressing:** Manual Static IP configuration and subnetting.
- **Backup Engine:** [rsync](#) (simulated) for automated daily data synchronization.

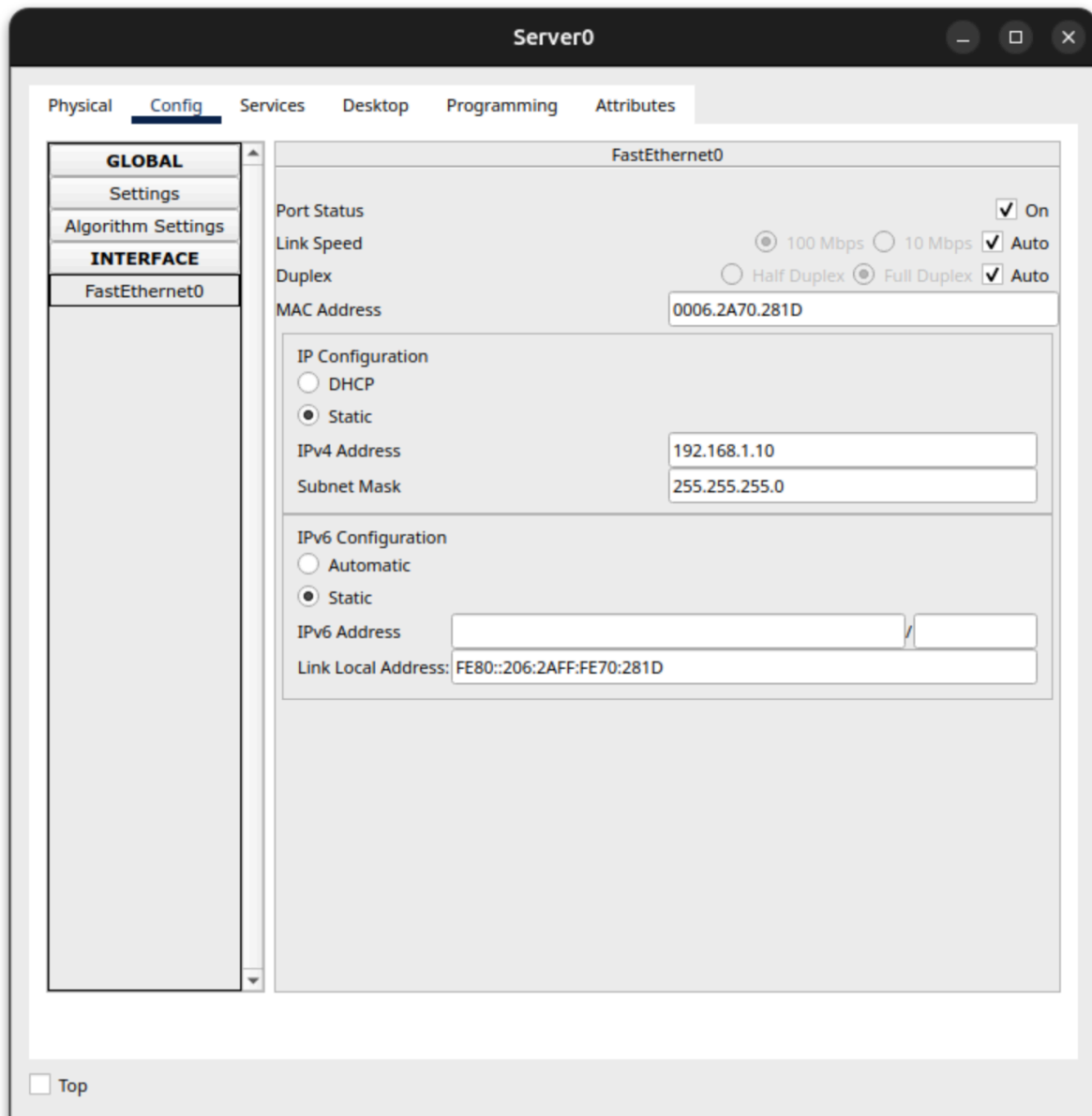


3. Configuration & Infrastructure

3.1 Network Addressing (IP Table)

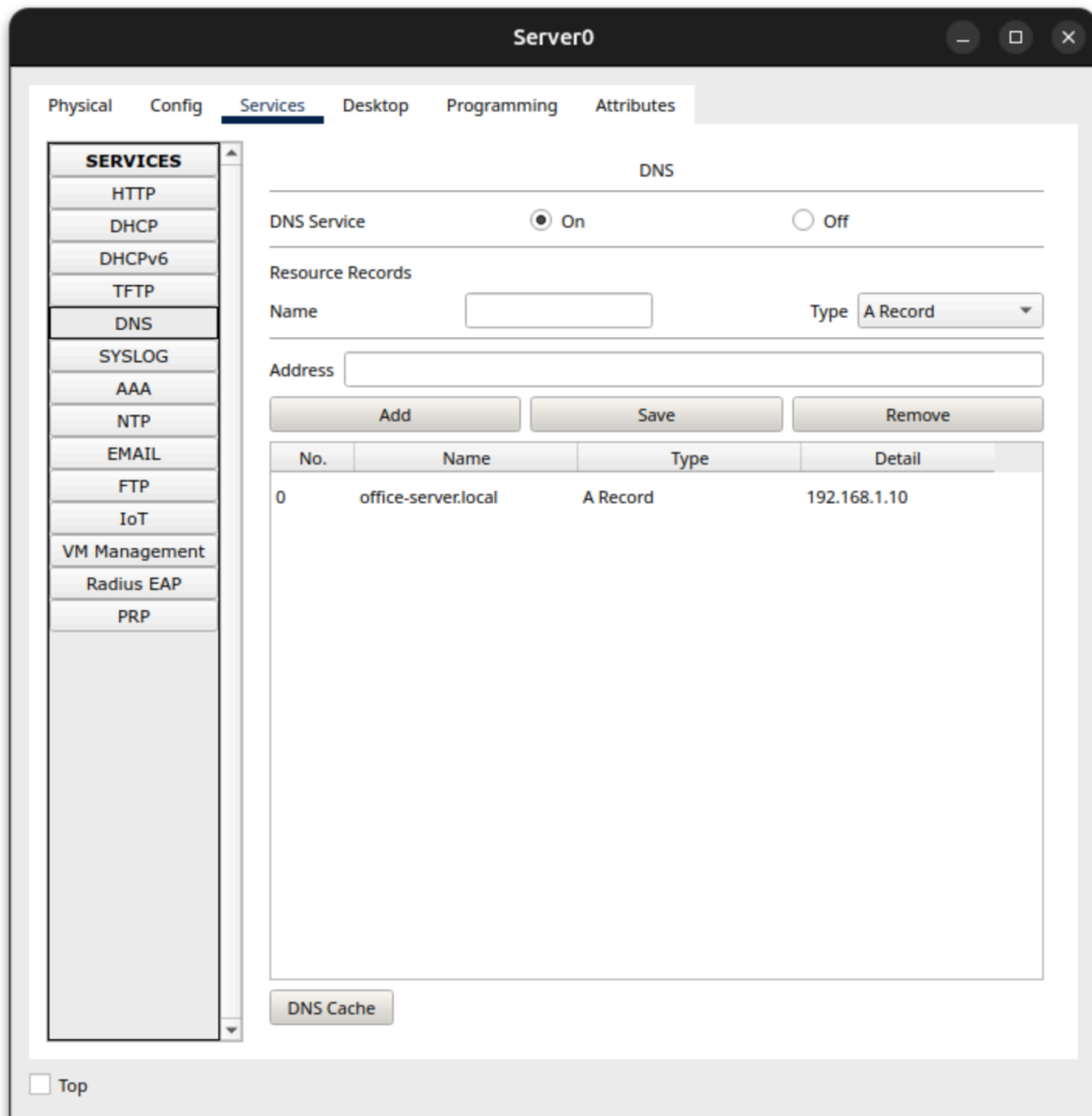
A static IP plan was implemented to ensure persistent connectivity across the local area network:

Device	IP Address	Subnet Mask	Role
Server	192.168.1.10	255.255.255.0	DNS, FTP, Backup
PC 0	192.168.1.11	255.255.255.0	Workstation
PC 1	192.168.1.12	255.255.255.0	Workstation
PC 2	192.168.1.13	255.255.255.0	Workstation
Printer	192.168.1.20	255.255.255.0	Network Printing



3.2 Services & Domain Resolution

- **Local DNS:** A local record was created to map `office-server.local` to `192.168.1.10`, allowing name-based access.
- **File Server (Samba/FTP):** Centralized storage was simulated via the FTP service on the server, providing role-based access for file management.



4. Backup Strategy & Validation

4.1 Automated Daily Backups

To prevent data loss, an automatic backup script was documented to run daily via **Cron**:

- **Mechanism:** `rsync -avz --delete /home/data/ 192.168.1.10:/backups/`
- **Frequency:** Every day at 00:00.

4.2 Failure & Recovery Simulation

The resilience of the network was verified through a loss-and-restore test:

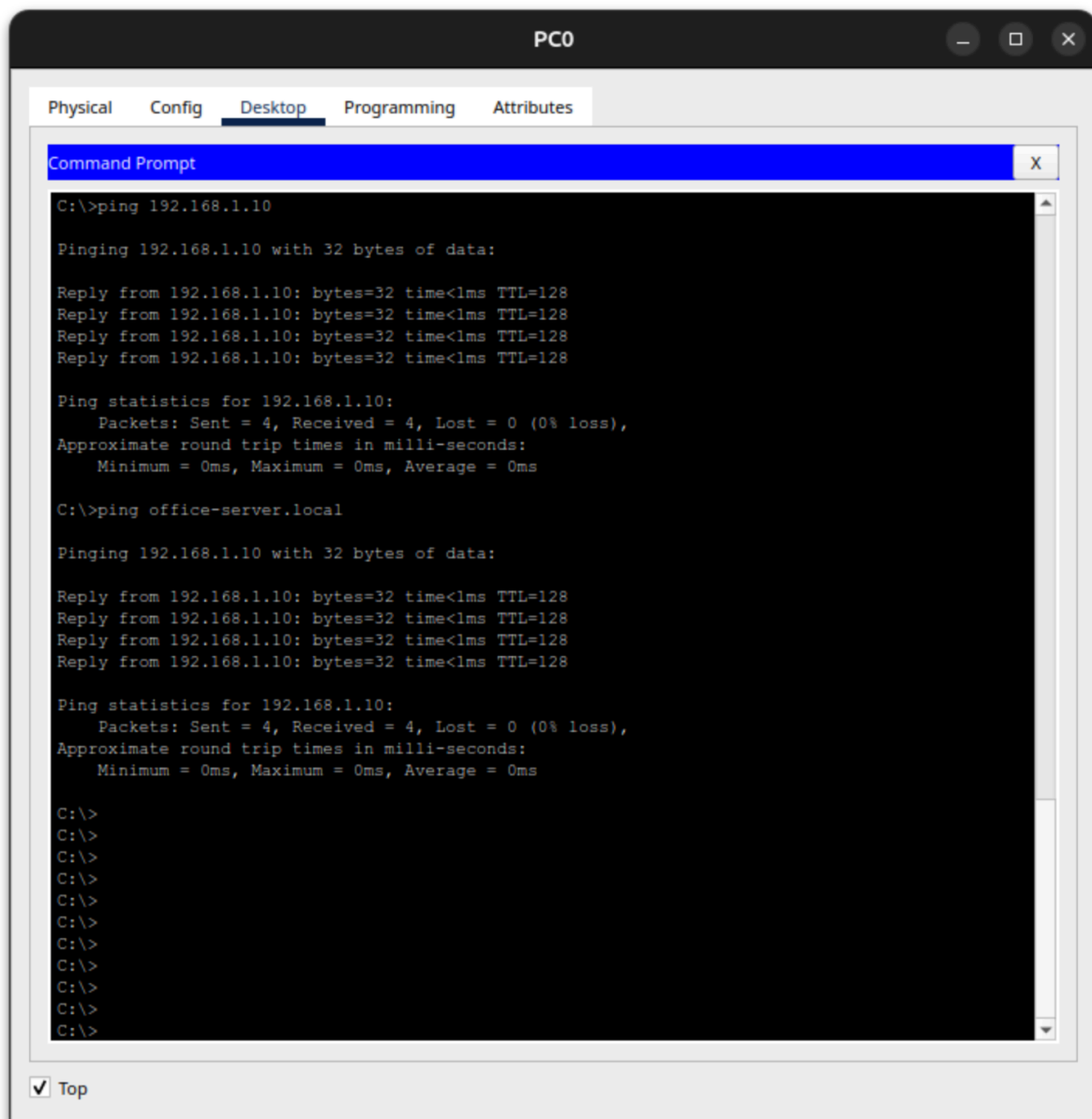
1. **Failure:** A test file was manually deleted from Client PC 0.
2. **Recovery:** The file was successfully restored by retrieving it from the Server's backup directory via the network.

5. System Validation (Proofs)

5.1 Connectivity Test

Connectivity between clients and the server was verified using the ICMP protocol.

- **Command:** `ping office-server.local`
- **Result:** Successful resolution and 0% packet loss.



The screenshot shows a PC0 desktop environment with a window titled "PC0". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Desktop" selected. Inside the window is a "Command Prompt" window. The Command Prompt shows the following text:

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
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Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

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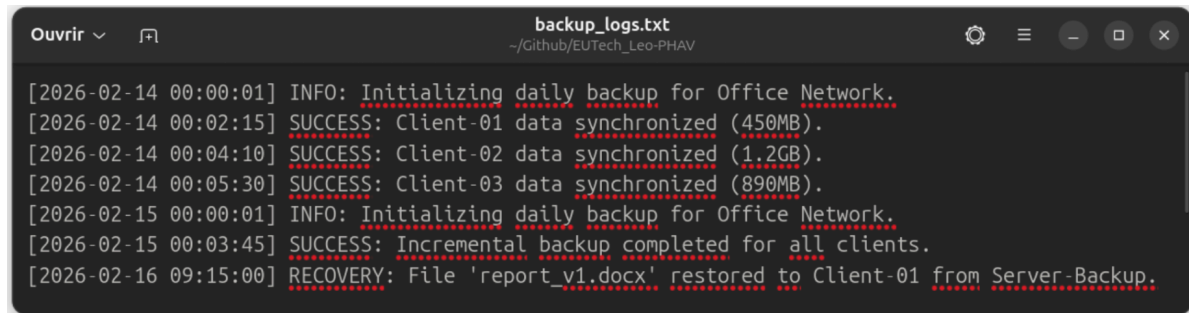
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```

At the bottom left of the Command Prompt window, there is a checkbox labeled "Top" which is checked.

5.2 Backup Logs

A log file was maintained to verify successful daily operations:



The screenshot shows a terminal window with the title bar 'backup_logs.txt' and the path '~/Github/EUTech_Leo-PHAV'. The window contains the following log entries:

```
[2026-02-14 00:00:01] INFO: Initializing daily backup for Office Network.
[2026-02-14 00:02:15] SUCCESS: Client-01 data synchronized (450MB).
[2026-02-14 00:04:10] SUCCESS: Client-02 data synchronized (1.2GB).
[2026-02-14 00:05:30] SUCCESS: Client-03 data synchronized (890MB).
[2026-02-15 00:00:01] INFO: Initializing daily backup for Office Network.
[2026-02-15 00:03:45] SUCCESS: Incremental backup completed for all clients.
[2026-02-16 09:15:00] RECOVERY: File 'report_v1.docx' restored to Client-01 from Server-Backup.
```